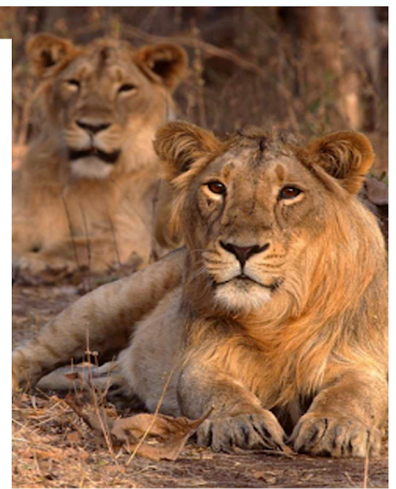
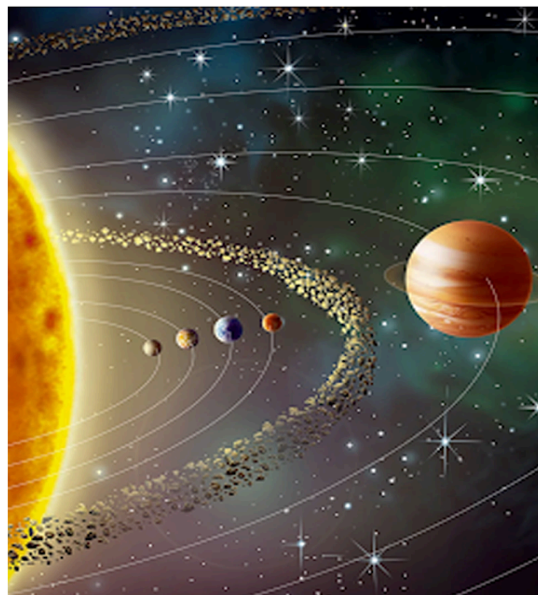
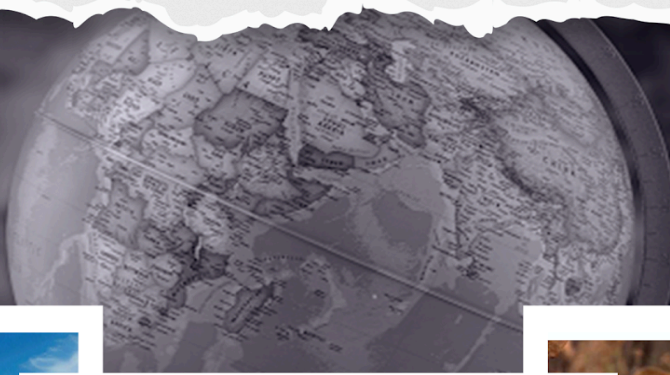


General Knowledge

World Geography

STUDY NOTES

Oceanography: Oceans

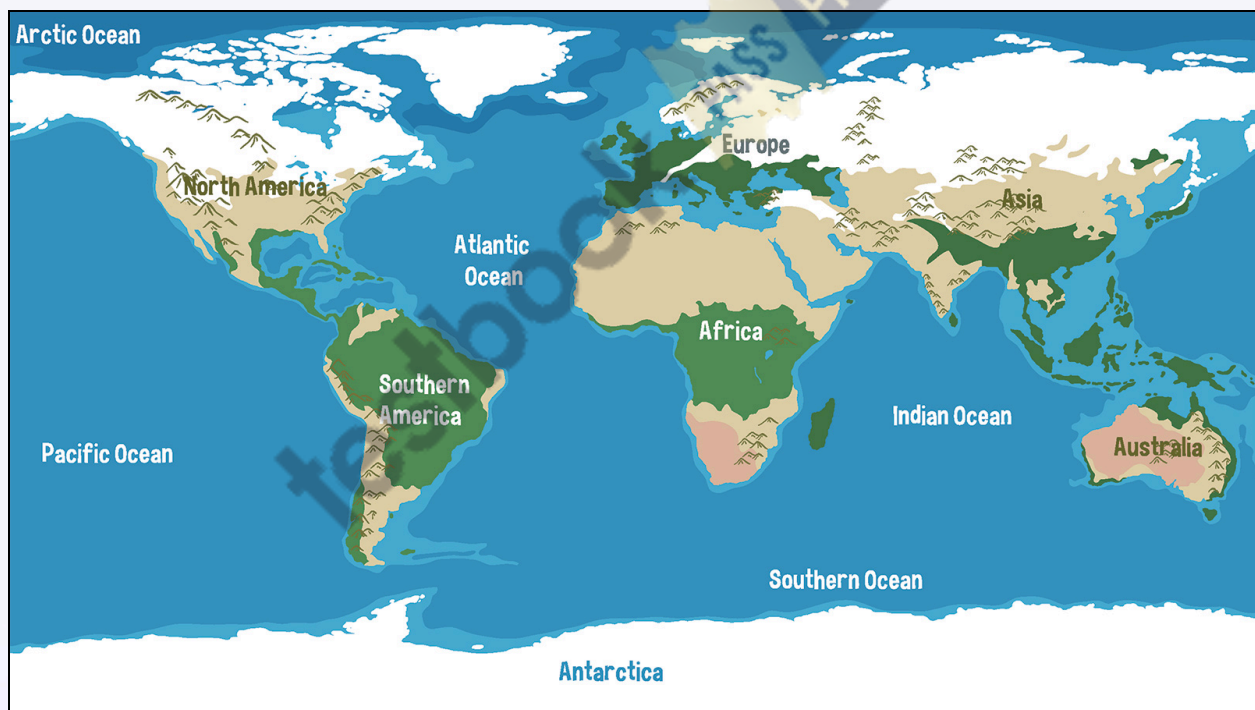


Oceanography: Oceans

Oceanography is a scientific discipline concerned with all aspects of the world's oceans and seas, including their physical and chemical properties, their origin and geologic framework, and the life forms that inhabit the marine environment.

The oceans are confined to the great depressions of the earth's outer layer. The oceans, unlike the continents, merge so naturally into one another that it is hard to demarcate them. The term **ocean is derived from the Greek word "Oceanus"** which means enormous river encircling the earth.

- The geographers have divided the oceanic part of the earth into five oceans, **namely the Pacific, the Atlantic, the Indian, the Southern Ocean, and the Arctic.**
- **The various seas, bays, gulfs, and other inlets are parts of these four large oceans.**



Pacific Ocean

- **Largest ocean** on Earth, covering **one-third** of the planet's surface—larger than all continents combined.
- Separates **Asia** and **Australia** from **North** and **South America**.

- Named "**Pacific**" by explorer **Ferdinand Magellan**, meaning "peaceful" due to its calm appearance.
- Home to the **International Date Line**.
- Connected to the **Indian Ocean** via the **Indonesian Seaway**, **Torres Strait**, and **Strait of Malacca**.
- Average depth: **4,280 meters**.
- Contains the **Mariana Trench**, the world's deepest oceanic trench.
- The marginal seas of the Pacific Ocean are **the Philippine Sea, Coral Sea, South China Sea, Tasman Sea, Bering Sea, East China Sea, Sea of Okhotsk, Sea of Japan, Yellow Sea, Celebes Sea, Sulu Sea, and the Sea of Chiloé**.



Atlantic Ocean

- The **second-largest ocean**, covering **20%** of Earth's surface.
- Named after **Atlantis Thalassa** (Sea of Atlas) from **Greek mythology**.
- Located between **Africa, Europe, the Arctic Ocean, the Americas, and the Southern Ocean**.

- The Atlantic Ocean is shaped like the letter "S".
- Average depth: **3,646 meters**.
- The **Puerto Rico Trench** is its deepest point at **8,376 meters**.
- Formed after the breakup of **Pangaea** (North Atlantic) and **Gondwanaland** (South Atlantic).
- **Greenland**, the world's largest island, is situated in the Atlantic.
- Home to the **Sargasso Sea**, the only sea without a shore.
- The **Atlantic** and **Indian Oceans** meet in the southeast.
- The marginal seas of the Atlantic Ocean are **the Caribbean Sea, Mediterranean Sea, Hudson Bay, Norwegian Sea, Greenland Sea, Scotia Sea, North Sea, Baltic Sea, Irish Sea, and the English Channel**.



Indian Ocean

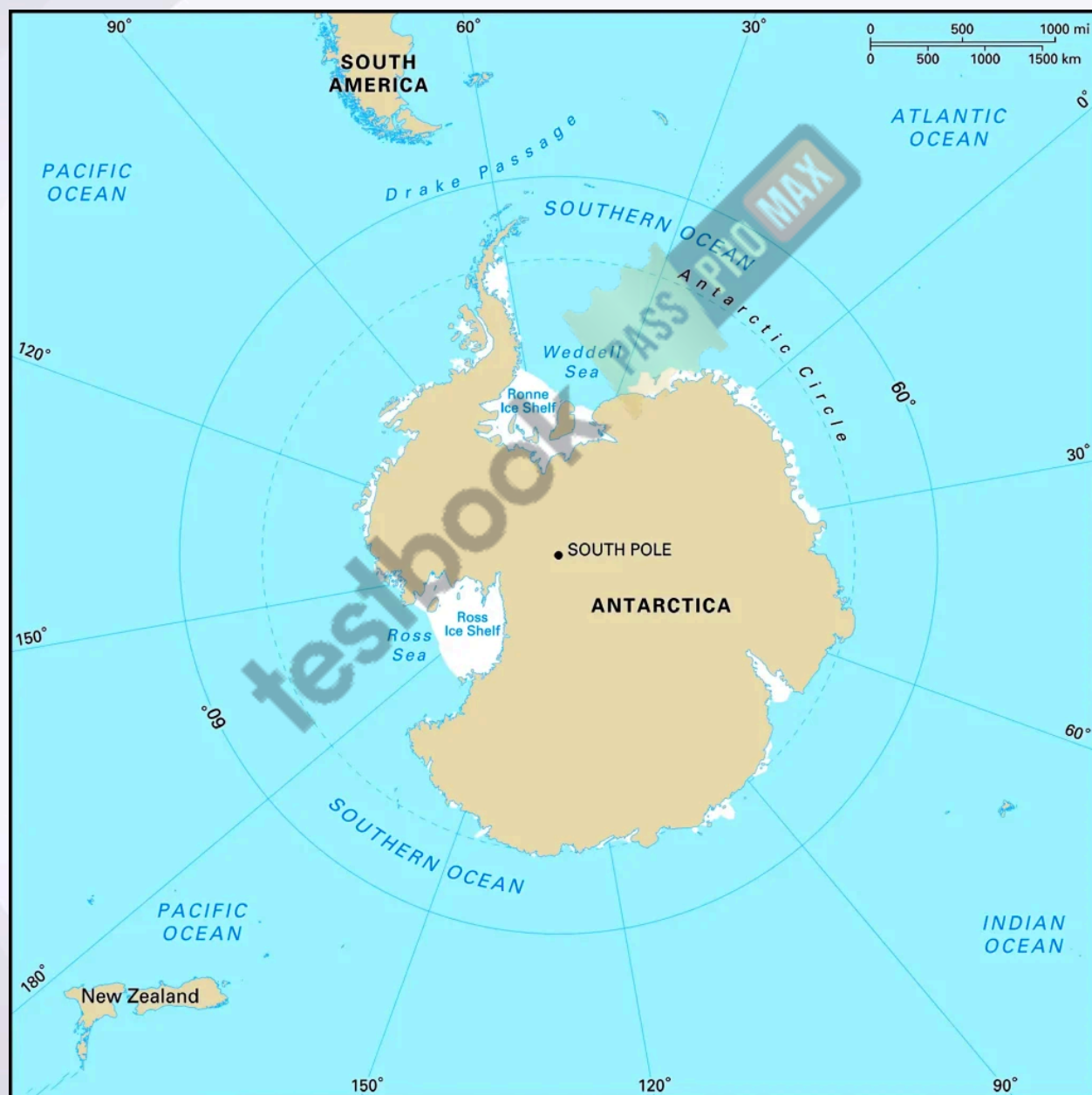
- Named after **India**, which forms much of its northern boundary.
- Located in the **Eastern Hemisphere**, covering **19.8%** of Earth's water.
- Bordered by **Asia**, **Australia**, **Africa**, and the **Southern Ocean**.
- Known as the world's **hottest ocean** and is surrounded by land in the northern hemisphere.
- **Madagascar** is its largest island.
- The **Sunda/Java Trench** is the deepest point at **7,450 meters**.
- Crucial for global **trade**, with marginal seas linking **East Asia**, **Africa**, **Europe**, the **Middle East**, and the **Americas**.
- The Marginal seas are the **Arabian Sea**, **Bay of Bengal**, **Andaman Sea**, **Red Sea**, **Java Sea**, **Persian Gulf**, and the **Sea of Zanj**.



Antarctic Ocean

- Also called the **Antarctic** or **Austral Ocean**, located around **Antarctica** south of **60°S latitude**.

- The **second-smallest**, but **fourth-largest ocean** in the world.
- Bordered by the **Atlantic, Indian, and Pacific Oceans** to the north.
- Known for **stormy seas** and extreme conditions.
- **The Drake Passage** between **Chile's Cape Horn** and the **South Shetland Islands** connects it to the southern Atlantic and Pacific Oceans.
- The marginal seas near the Antarctic Ocean are the **Amundsen Sea, Bellingshausen Sea, Weddell Sea**, and the **Ross Sea**.



Arctic Ocean

- The **smallest** and **shallowest** of the five oceans, covering **1.3%** of Earth's water.
- Encompasses the **Arctic Circle** and is fully covered by **ice** in winter.
- Bounded by **Eurasia** and **North America**.
- Has the **lowest salinity** of all the oceans.
- Connected to the **Pacific** through the **Bering Strait** and to the **Atlantic** by the **Greenland Sea**, **Labrador Sea**, and the **Norwegian Sea**.
- The **Molloy Deep** in the **Fram Strait** is its deepest point.
- The Marginal seas are **the Barents Sea** and **the Irish Sea**.

